14. Sentence Agreement Checking Using CFG Rules

Aim: To check subject-verb agreement using simple context-free grammar rules in Python.

Procedure:

1. Import NLTK.
2. Define grammar rules for singular and plural subjects.
3. Create a parser.
4. Test sentences with matching and mismatching subjects and verbs.
5. Display whether each sentence is accepted.

Code:

```
import nltk
grammar = nltk.CFG.fromstring("""
S -> NS VS | NP VP
NS -> 'he' | 'she'
VS -> 'runs' | 'plays'
NP -> 'they' | 'we'
VP -> 'run' | 'play'""")
parser = nltk.ChartParser(grammar)
sentences = [
    "he runs",
    "they run",
    "he run",
    "they runs"
]
for sentence in sentences:
    try:
 next(parser.parse(sentence.split()))
        print(sentence, "- Accepted")
```

```
    except StopIteration:
        print(sentence, "- Rejected")
```

Output:

```
C:\Users\blohi\Downloads\NLP lab>C:\Users\blohi\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/b
lohi/Downloads/NLP lab/14.py"
he runs - Accepted
they run - Accepted
he run - Rejected
they runs - Rejected
```

Result: Subject-verb agreement was checked using grammar rules.